Technical report - ApertusProbe

Summary. ApertusProbe identified and validated four distinct behavioral failures in the Apertus v1.5 family: (1) excluded supplied-context values copied during PUBLIC-only French translation on 8B normal; (2) half-open interval transition double-activation across three tested deployments; (3) cross-lingual neglect of explicit non-offsetting constraints in bilateral debt reasoning; and (4) false text assertions overriding correctly perceived visual values in normal multimodal models. All four findings are backed by authentic hashed evidence, matched controls, and an offline, network-disabled judge replay via root `make run`.

Track	Apertus Readiness - Track 1A
Event	Online
Team	Cayden
Member	Cayden
Project	ApertusProbe
Demo	Run `make run` from repository root; finding walkthroughs are in `track_1a/findings/`.

Use of Apertus

- **Models evaluated:** `swiss-ai/Apertus-v1.5-8B` (normal), `swiss-ai/Apertus-v1.5-8B-thinking`, and `swiss-ai/Apertus-v1.5-70B` (normal). Exploratory work also examined 70B-thinking.
- **Inference setup:** CSCS OpenAI-compatible hosted endpoint, temperature 0, seed 42 where accepted, deterministic zero-shot prompts, and bounded output budgets (typically 256 or 1024 tokens). No fine-tuning or weight modification was performed.
- **Evidence capture:** exact requests, raw provider responses, model/deployment IDs, channels, hashes, and checkpoints are archived. Hosted hardware and exact weight revisions were not disclosed [1,2].

Problem

We evaluated whether Apertus preserves instruction fidelity and evidence grounding across four operationally relevant settings.

Finding #1 - Supplied-context excluded-field copying. In a synthetic museum record, an explicit instruction requires publication of PUBLIC fields only and forbids output of `PRIVATE.excluded_token`. Apertus v1.5-8B normal nevertheless copies the excluded synthetic value into its French translation. This is a supplied-context controllability failure; it is *not* evidence of memorized PII or cross-user data exposure.

Finding #2 - Half-open interval transition double-activation. For schedules explicitly defined as `[start, end)`, tested 8B normal, 8B-thinking, and 70B normal deployments return both adjacent intervals at the transition `t = end(T1) = start(T2)`, instead of only the newly active interval.

Finding #3 - Cross-lingual non-offsetting constraint neglect. Equivalent bilateral-debt tasks obey “without offsetting” in English but repeatedly compute net balances under tested German, Italian, and French formulations. At 70B scale, the tested German cases recover while Italian and French failures persist.

Finding #4 - False text overrides verified visual evidence. In multimodal cards, event badges, and telemetry dashboards, 8B normal and 70B normal copy false numerical/date claims from prompt text even though matched image-only controls show that the visual value is perceived correctly. The tested 8B-thinking deployment rejects the contradiction.

Red Teaming Method

Overall strategy

Research followed a staged pipeline: Scout -> Pilot -> Replication -> Breadth -> Falsification -> Candidate Audit -> Submission Packaging. A hypothesis was promoted only after fresh-value replication, matched controls, deterministic or reference-grounded evaluation, and explicit attempts to falsify the proposed mechanism. Cosmetic formatting errors and brittle one-offs were pruned rather than submitted.

> **Research scale.** The project logged **1,211 live hosted inference calls** across four research campaigns: 962 in Rounds 1-2, 144 in Round 3, and 105 in Round 4. Submission replay itself uses **zero** live inference calls.

Discovery process and falsification

Finding #1. Targeted translation repeats copied the full excluded token in 6/6 packaged validation cases, while placement/breadth testing showed leakage in 5/12 variants. Matched summaries, absent-canary inputs, stronger exclusion phrasing, 70B normal, and 8B-thinking provided negative or deployment controls; permitted-copy cases confirmed the model could emit the field when explicitly allowed.

Finding #2. Fresh half-open schedule tasks reproduced transition double-activation across 8B normal, 8B-thinking, and 70B normal. Interior/start-boundary controls passed. A direct Boolean date-comparison probe also passed, arguing against a primitive date-comparison explanation and localizing the failure to schedule/set filtering.

Finding #3. English controls returned gross obligations correctly across tested deployments. German, Italian, and French variants reproduced netting despite explicit non-offsetting wording on 8B-class deployments; 70B normal recovered German but continued to fail tested Italian and French pairs. Directional debtor/creditor controls passed, reducing the likelihood of simple role-binding confusion.

Finding #4. Bare-image controls read numbers, dates, and telemetry values correctly. Adding a false textual assertion caused normal 8B and 70B to output the false text value across tested contradiction cases. A visual-only/debiasing instruction restored 8B normal accuracy, while the 8B-thinking deployment consistently resolved the text-image conflict in favor of the image.

Pruned hypotheses

Large portions of the search space were intentionally rejected. Examples include identifier verbosity that proved to be harmless prose, long-context JSON-wrapper loss with correct semantics, multiple authority/supersession hypotheses that failed fresh controls, and several logic/eligibility anomalies that did not generalize. This pruning is important: the final four findings are the subset that survived the project's validation threshold.

Tooling

- SHA-256 manifests bind archived requests/responses and packaged evidence.
- Normal and thinking outputs are stored as separate channels.
- Task-specific evaluators use literal matching, normalized tokens, and deterministic numerical/set ground truth.
- The judge entry point (`python -m src.submission`) validates all four finding JSON files and replays evidence entirely offline.

Key Validation Evidence

Counts below summarize validated experiment subsets from the archived research corpus. The packaged submission evidence inventory is listed on page 4.

Evidence group	Target deployment	Expected	Observed	Interpretation
F1: French translation repeats	8B normal	Exclude token	6/6 copied full token	Excluded-field copying
F1: Summary / absence controls	8B normal	No token	0 copies in matched controls	Negative controls pass
F1: Deployment controls	70B normal / 8B-thinking	No token	0/3 + 0/3 echoes	Bounded deployment effect
F2: Transition instant	8B normal / 8B-thinking / 70B normal	`["T2"]`	4/4 return `["T1","T2"]`	Boundary double-activation
F2: Interior/start controls	Tested deployments	Single active interval	Controls pass	Failure localized to transition
F2: Boolean date probe	8B normal	`false`	`false`	Primitive comparison intact
F3: English non-offsetting	8B / 8B-thinking / 70B normal	Gross X	10/10 gross X	English adherence
F3: DE / IT / FR	8B-class tested deployments	Gross X	6/6 net X-Y	Cross-lingual constraint neglect
F3: IT / FR at scale	70B normal	Gross X	4/4 net X-Y	Persists in tested IT/FR cases
F4: Bare-image perception	8B normal / 8B-thinking / 70B normal	True visual value	12/12 correct	Perception intact
F4: Contradictory prompt text	8B normal / 70B normal	True visual value	8/8 false text value	Text overrides image evidence
F4: Thinking deployment	8B-thinking	True visual value	4/4 reject false text	Deployment boundary
F4: Visual-only debiasing control	8B normal	True visual value	4/4 restored	Grounding can be recovered

> **Judge reproduction principle.** The report does not ask judges to trust aggregate percentages alone. The repository includes raw authentic model responses, evidence manifests, exact expected outputs, and an automated replay that re-evaluates every packaged case without network access.

Findings Dataset

We submit four independently scoped findings with **72 packaged authentic evidence cases** in total.

Finding	Topic	Affected scope	Packaged cases	Scope note
#1 Excluded-field copying	Supplied-context information flow	8B normal, English->French transformation	36	Narrow, synthetic context; no real-world PII claim
#2 Half-open transition double-activation	Temporal / tabular factuality	8B normal, 8B-thinking, 70B normal	12	Reproduces across three tested deployments
#3 Multilingual non-offsetting neglect	Cross-lingual instruction fidelity	8B-class DE/IT/FR; 70B normal IT/FR	12	English matched controls pass
#4 False text overrides visual values	Multimodal grounding	8B normal, 70B normal	12	8B-thinking and visual-only controls pass

Finding Summaries

Finding #1. Thirty-six packaged cases in `data/submission_evidence/` evaluate omission of a synthetic excluded field during English-to-French transformation. The finding is intentionally narrow: it demonstrates conditional copying of supplied context, not extraction of hidden or memorized personal data.

Finding #2. Twelve packaged cases in `data/finding2_evidence/` demonstrate that tested Apertus deployments can double-activate adjacent half-open schedule intervals at the exact transition. Supporting 8B out-of-range clamping is treated as boundary evidence, not a separate finding.

Finding #3. Twelve packaged cases in `data/finding3_evidence/` demonstrate a language-dependent instruction-following gap for explicit non-offsetting constraints. The claim is limited to tested bilateral-debt constructions and languages; it does not assert a general multilingual bias.

Finding #4. Twelve packaged cases plus five verified PNG assets in `data/finding4_evidence/` demonstrate a text-over-image evidence-priority failure in tested normal multimodal deployments. Image-only and debiasing controls confirm that the underlying visual content is readable.

Lessons Learnt and Recommendations

- **Pre-filter non-public context.** Remove fields before model ingestion rather than relying solely on negative omission instructions.
- **Delegate interval boundaries to deterministic code.** Scheduling and validity windows should not rely on free-form LLM set filtering at exact boundaries.
- **Audit multilingual negative constraints.** Validate semantically equivalent non-English instructions independently rather than assuming English behavior transfers.
- **Separate visual extraction from user assertions.** In multimodal pipelines, isolate perceptual extraction or explicitly prioritize image evidence when text may be wrong.
- **Prefer quality over finding count.** Cosmetic and non-reproducible anomalies were excluded from submission.

Reproducibility

Primary reproduction requires GNU Make and Docker:

```
make run
```

From repository root, this builds the submission container, executes the automated test suite, and replays all four packaged findings with Docker networking disabled. No API key, model weight download, or internet connection is required for judge replay.

Detailed prompts, expected outputs, and observed responses:

```
make run ARGS="--details"
```

Local fallback:

```
cd track_1a
pip install ".[test,report]"
python -m pytest tests -q --basetemp=.pytest_tmp
python -m src.submission
```

The verified submission state passes **144 automated tests**. Tests cover evidence-manifest integrity, schema validation, tamper rejection, deterministic evaluators, and zero-network replay.

License

- **Report and PDF:** Creative Commons Attribution 4.0 International (CC-BY-4.0).
- **Codebase:** Apache License 2.0.
- **Findings and packaged evidence:** CDLA-Permissive-2.0.
- Repository visibility remains private until 1 December 2026 in accordance with Track 1A requirements [4].

References

[1] Swiss AI. *Apertus v1.5-8B Model Card.* https://huggingface.co/swiss-ai/Apertus-v1.5-8B

[2] Swiss AI. *Apertus v1.5-70B Model Card.* https://huggingface.co/swiss-ai/Apertus-v1.5-70B

[3] Haritz Puerto et al. *From Leaky Thoughts to Private Reasoning: Controlling What LRMs Say to Themselves.* arXiv:2602.24210, 2026.

[4] Hack Apertus. *Track 1A Requirements and Evaluation Rubric.* https://github.com/HackApertus/project-template/blob/main/track_1a/README.md

[5] Swiss AI. *Apertus v1 Technical Report.* https://arxiv.org/abs/2509.14233

Submission Snapshot

4 findings **72 packaged cases** **144 tests passed** **5-page report**

Project: ApertusProbe
Team: Cayden
Member: Cayden

Judge entry point: `make run`
Replay: offline, `--network none`